Abhishek Tiwari
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PROFESSIONAL SUMMARY

Senior Machine Learning Engineer with 4.5 years of production experience across SaaS and enterprise software. Deep hands-on expertise in PyTorch, TensorFlow, FastAPI, Docker. Known for measurable impact on latency, reliability and cost, and for mentoring engineers.

TECHNICAL SKILLS

PyTorch, TensorFlow, FastAPI, Docker, AWS, Hugging Face, MLOps, Scikit-Learn, Redis, MLflow, CI/CD, Kubernetes, Natural Language Processing, Pandas, Airflow

EDUCATION

M.Tech in Computer Science
IIT Madras | CGPA: 8.5/10 | Graduated: 2021

WORK EXPERIENCE

Senior Machine Learning Engineer | Copperleaf Payments (Jul 2026 - Jan 2024)
• Built an MLOps platform on MLflow, Airflow and Docker with automated retraining and canary rollout, cutting model release time from 12 days to 3 hours.
• Deployed RoBERTa ONNX models with FastAPI and Triton serving 95k daily requests at 38 ms p95 latency on AWS ECS.
• Reduced AWS GPU spend by 32% (about $14k per month) through INT8 quantization, dynamic batching and spot-instance training.
• Led a squad of 6 engineers, introduced PyTest-based model contracts and lifted coverage of the training stack to 91%.

Machine Learning Engineer | Basalt Mobility (Apr 2022 - May 2024)
• Designed a feature store on PostgreSQL and Redis with point-in-time correctness; online feature reads at 4 ms p99.
• Shipped a TensorFlow Serving to PyTorch migration for 14 legacy models with zero downtime and a 18% latency improvement.
• Raised macro-F1 from 0.78 to 0.94 on a 24-class ticket classifier by fine-tuning Hugging Face transformers with focal loss and error-driven data curation.

KEY PROJECTS

Open-Source: onnx-batching-utils
• Maintainer of a PyTorch-to-ONNX export and dynamic-batching helper (pytorch + FastAPI + Triton) with 1400 GitHub stars and 6 external contributors.

Semantic Resume Search
• RAG pipeline using sentence-transformers, FAISS and a FastAPI backend; returns results in under 62 ms across 120,000 documents.

CERTIFICATIONS & AWARDS

• TensorFlow Developer Certificate
• Runner-up, Google Cloud Hackathon India 2022